Cloud-Based Business Data Analytics and Decision Support System

Project Report

Submitted by: Baisakhi Bhaduri

Roll Number: O2BCA110280

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Abstract

This project analyses DataServe India and benchmarks it against Competitor Alpha and Competitor Beta using the supplied Excel datasets. Python/Pandas calculates KPIs and group summaries; Streamlit provides filters, comparison tables, visualisations and CSV export.

1. Introduction

Cloud analytics decisions require a clear view of query performance, availability, costs and data quality. This project turns spreadsheet data into comparable indicators and recommendations.

2. Problem Statement

The task is to identify performance gaps, examine patterns across month, region, mining segment and service model, and prioritise improvements using measurable KPIs.

3. Objectives

• Inspect all three datasets and audit basic data quality.

• Calculate comparable KPIs consistently.

• Analyse DataServe by month, region, segment and service model.

• Build an interactive dashboard and generate recommendations.

4. Dataset Description

Dataset	Records	Fields	Role
DataServe India	1600	20	Client analysis
Competitor Alpha	1100	20	Benchmark
Competitor Beta	1100	20	Benchmark

The 20 shared source fields include month, region, business unit, customer segment, cloud service model, data volume, query count/time, cloud cost, uptime, dashboard usage, data quality, mining segment, conversion, satisfaction, optimisation, storage utilisation and decision effectiveness.

5. Methodology

Datasets were loaded from their Dataset worksheets, exported to CSV for portability, audited for missing cells and duplicate record IDs, then aggregated. Average KPI scores use arithmetic means. Cost per GB is total cloud cost divided by total data volume. The dashboard groups records by month, region, mining segment and service model.

6. Technology Stack

Component	Technology	Purpose
Data processing	Python, Pandas, NumPy	Load and aggregate datasets
Dashboard	Streamlit	Interactive filters and KPI views
Charts	Matplotlib / Streamlit	Comparisons and grouped trends
Input	Provided Excel files / bundled CSV	Reproducible local analysis

7. KPI Definitions

KPI	Definition
Average query time	Mean of Avg_Query_Time_Sec

KPI	Definition
Average uptime	Mean of Uptime_Pct
Cost per GB	Sum(Cloud_Cost_INR) / Sum(Data_Volume_GB)
Data quality	Mean of Data_Quality_Score
Decision effectiveness	Mean of Decision_Effectiveness_Score

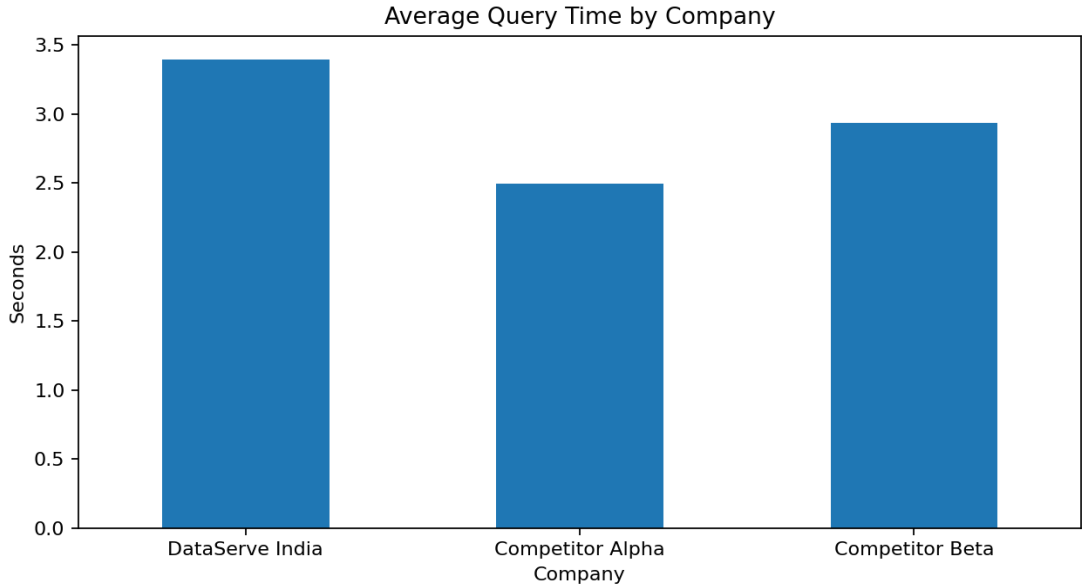
8. Company Benchmark Results

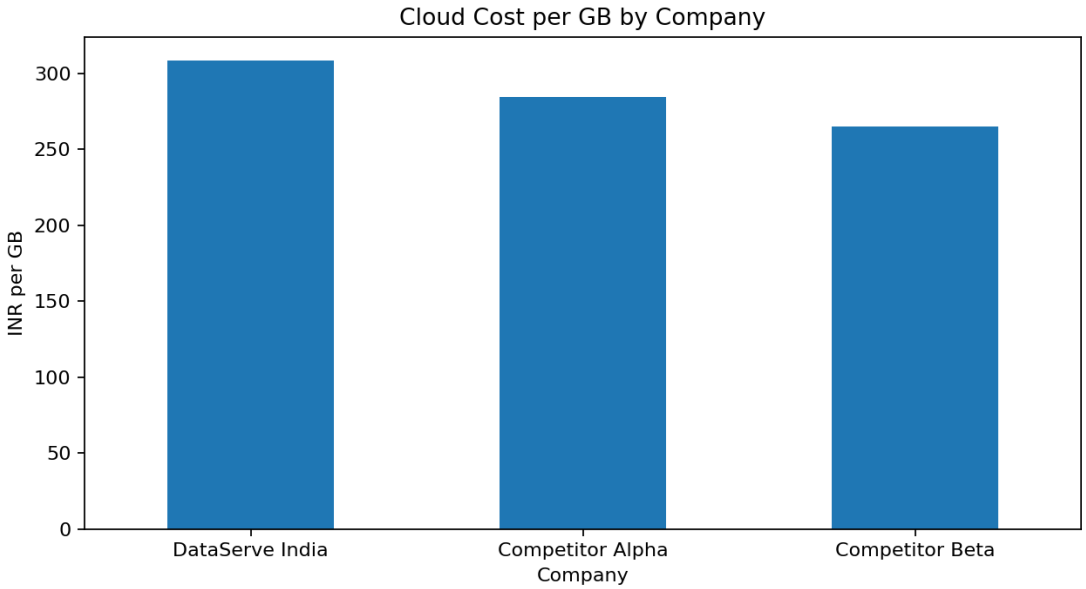
Company	Records	Query time (s)	Uptime (%)	Cost/GB	Data quality	Conversion (%)	Decision effect.
DataServe India	1600	3.40	97.39	■308.63	74.78	11.89	74.30
Competitor Alpha	1100	2.50	98.52	■284.63	83.19	15.02	85.60
Competitor Beta	1100	2.94	98.97	■265.16	80.43	13.52	82.78

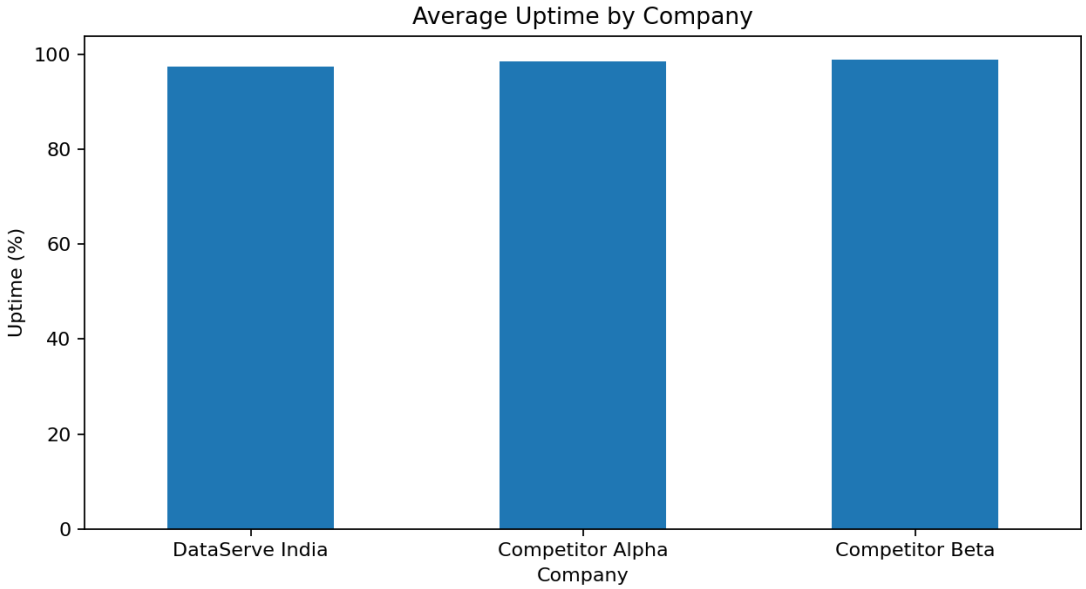
DataServe averages 3.40s query time, 97.39% uptime, ■308.63/GB, data quality 74.78 and decision effectiveness 74.30. Alpha is fastest at 2.50s; Beta has highest uptime (98.97%) and lowest cost/GB (■265.16).

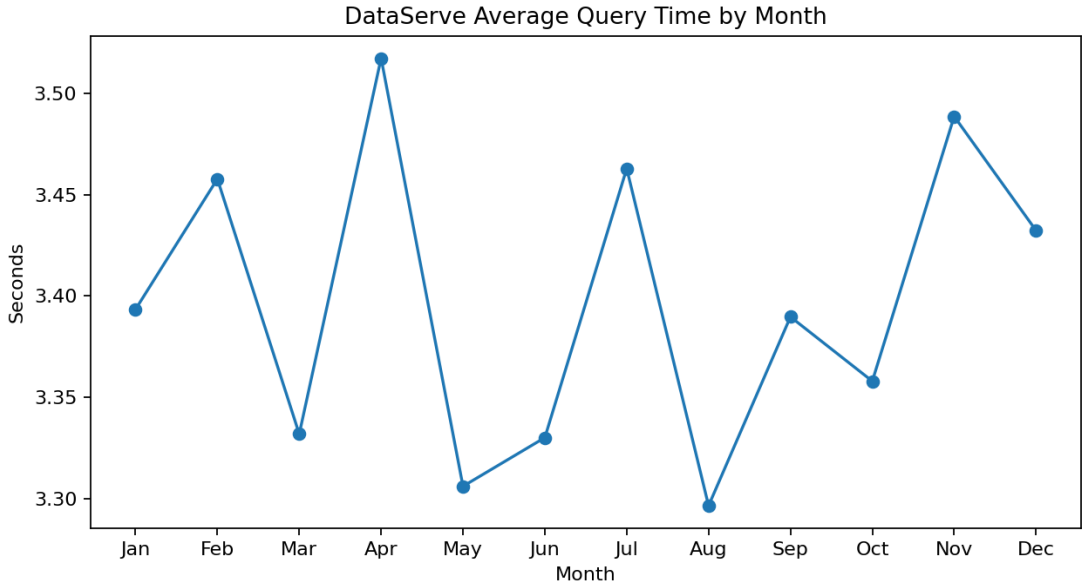
9. DataServe Monthly Analysis

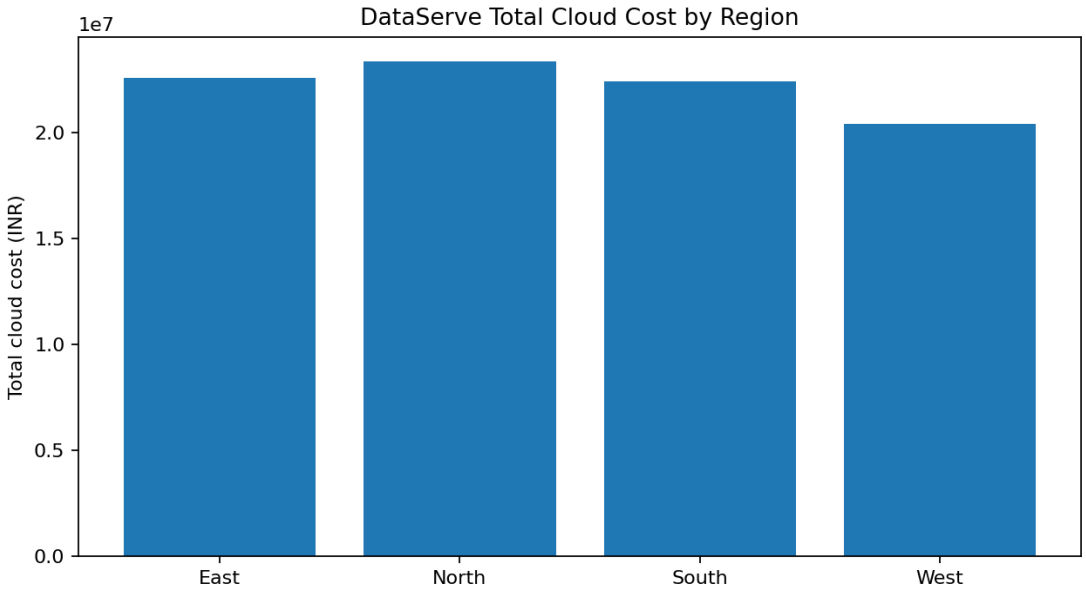
Month	Records	Avg query (s)	Avg cost (■)	Data quality	Conversion (%)
Jan	124	3.39	55451	75.13	11.66
Feb	136	3.46	56949	74.69	11.72
Mar	136	3.33	54422	74.49	12.26
Apr	123	3.52	54111	73.59	11.37
May	121	3.31	55578	76.20	11.88
Jun	147	3.33	56227	73.89	11.90
Jul	127	3.46	56231	74.78	11.82
Aug	141	3.30	55968	75.28	11.87
Sep	135	3.39	54421	74.43	11.76
Oct	131	3.36	54483	76.08	11.97
Nov	128	3.49	56181	74.56	12.06
Dec	151	3.43	55055	74.46	12.34

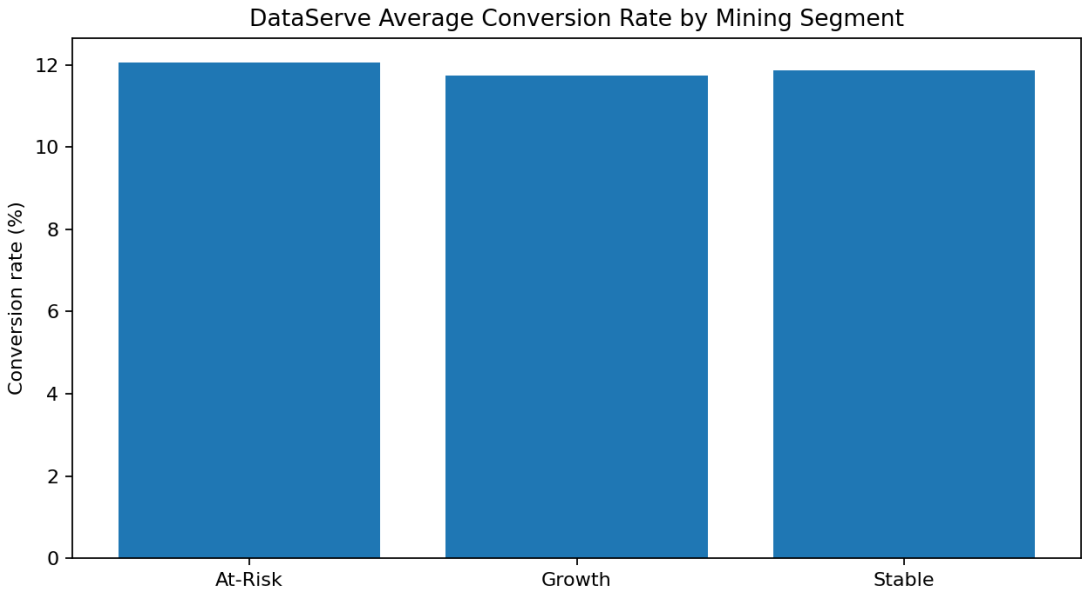

Average Query Time by Company
Seconds
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
DataServe India
Competitor Alpha
Competitor Beta
Company


Cloud Cost per GB by Company
INR per GB
300
250
200
150
100
50
0
DataServe India
Competitor Alpha
Competitor Beta
Company


Average Uptime by Company
Uptime (%)
100
80
60
40
20
0
DataServe India
Competitor Alpha
Competitor Beta
Company







10. Data Quality Audit

Dataset	Rows	Fields	Duplicate IDs	Missing cells	Months	Regions
DataServe India	1600	20	0	0	12	4
Competitor Alpha	1100	20	0	0	12	4
Competitor Beta	1100	20	0	0	12	4

The audit describes the supplied files only. A low missing/duplicate count does not prove that values are semantically valid.

11. Findings

• Query performance gap: DataServe averages 3.40 seconds, slower than Alpha (2.50 seconds) and Beta (2.94 seconds).

• Cost efficiency gap: DataServe's cost per GB is ■308.63, compared with Alpha ■284.63 and Beta ■265.16.

• Reliability gap: DataServe averages 97.39% uptime, below Alpha (98.52%) and Beta (98.97%).

• Data quality gap: DataServe's average score is 74.78, below Alpha (83.19) and Beta (80.43).

• Decision effectiveness: DataServe averages 74.30, below Alpha (85.60) and Beta (82.78).

12. Recommendations

Priority	Recommendation	Target
Query speed	Profile slow queries; test indexing, caching, simplification and workload scheduling.	Average query time toward ≤2.80 s.
Cloud cost	Review idle compute, storage tiers and workload cost.	Cost/GB toward ≤■284.63 (Alpha reference).
Reliability	Review incidents, availability patterns and recovery.	Uptime toward ≥98.5%.
Data quality	Validation, duplicate detection, data ownership and regular reports.	Data-quality score toward ≥80.
Decision support	Use dashboard in monthly reviews and monitor segment results.	Track conversion and decision effectiveness.

Targets are proposed goals, not achieved results. Pilot changes and use comparable workloads before attributing improvement to a particular intervention.

13. Limitations

• Datasets have different record counts and may have different workload mixes.

• Cost per GB is not a full cost-efficiency model.

• Averages may hide outliers.

• Observational comparisons do not prove causation.

• No live cloud billing or infrastructure connection is included.

14. Conclusion

The project converts the supplied spreadsheets into comparable KPIs and an interactive dashboard. DataServe has higher query time and cost per GB and lower uptime, data quality and decision effectiveness than both competitors. Monthly, regional and segment views help users explore these gaps and define measurable improvement goals.

15. References

• Microsoft Power BI: https://learn.microsoft.com/en-us/power-bi/

• Google Cloud Architecture Framework: https://cloud.google.com/architecture/framework

• Microsoft Learn data mining: https://learn.microsoft.com/en-us/analysis-services/data-mining/data-mining-concepts